Linear Equations & Slope Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Slope formula

ORIGINAL PROBLEM

Find the slope through (2, 3) and (6, 11).

INCORRECT WORKED SOLUTION

$$m = \frac{\text{change in x}}{\text{change in y}}$$

$$m = \frac{6 - 2}{11 - 3}$$

$$m = \frac{4}{8} = \frac{1}{2}$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Order of subtraction

ORIGINAL PROBLEM

Find the slope through (1, 5) and (4, −1).

INCORRECT WORKED SOLUTION

$$m = \frac{y_2 - y_1}{x_1 - x_2}$$

$$m = \frac{-1 - 5}{1 - 4}$$

$$m = \frac{-6}{-3} = 2$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Reading slope-intercept form

ORIGINAL PROBLEM

Identify the slope and y-intercept of $y = 5 - 2x$.

INCORRECT WORKED SOLUTION

$y = 5 - 2x$

$y = mx + b$

The first number is the slope: $m = 5$

The second number is the y-intercept: $b = -2$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Solving for y

ORIGINAL PROBLEM

Write $3x + 2y = 12$ in slope-intercept form.

INCORRECT WORKED SOLUTION

$3x + 2y = 12$

$2y = 12 + 3x$

$y = 6 + \frac{3x}{2}$

$y = \frac{3}{2}x + 6$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Writing an equation

ORIGINAL PROBLEM

Write the equation of the line with slope 3 through (2, 5).

INCORRECT WORKED SOLUTION

$y = mx + b$

$y = 3x + b$

The line passes through (2, 5), so b = 5.

$y = 3x + 5$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Parallel lines

ORIGINAL PROBLEM

Write the line parallel to y = 2x + 1 through (4, 3).

INCORRECT WORKED SOLUTION

Parallel lines have opposite reciprocal slopes.

$m = -\frac{1}{2}$

$3 = -\frac{1}{2}(4) + b \rightarrow b = 5$

$y = -\frac{1}{2}x + 5$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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7. Perpendicular lines

ORIGINAL PROBLEM

Find the slope of a line perpendicular to $y = \frac{2}{3}x - 4$.

INCORRECT WORKED SOLUTION

The slope of the given line is $\frac{2}{3}$.

Perpendicular slopes are opposites.

$m = -\frac{2}{3}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Finding an intercept

ORIGINAL PROBLEM

Find the x-intercept of $y = 2x - 6$.

INCORRECT WORKED SOLUTION

$y = 2x - 6$

Let $x = 0$: $y = 2(0) - 6 = -6$

The x-intercept is -6.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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9. Vertical lines

ORIGINAL PROBLEM

Find the slope of the line through (3, 2) and (3, 7).

INCORRECT WORKED SOLUTION

$$m = \frac{7 - 2}{3 - 3}$$

$$m = \frac{5}{0}$$

$$m = 0$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Point-slope form

ORIGINAL PROBLEM

Write the equation of the line with slope 2 through $(-3, 4)$ in point-slope form.

INCORRECT WORKED SOLUTION

$$y - y_1 = m(x - x_1)$$

$$y - 4 = 2(x - 3)$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points